

# Machine Learning – Regression – Assignment

## 1.Simple Linear Regression and Multiple Linear Regression

R2 Value = 0.7894790349867009

## 2.Support Vector Machine

| S.NO | Hyper Parameter | Linear (R Value) | RBF (non-linear) (R Value) | Poly (R Value) | Sigmoid (R Value) |
|------|-----------------|------------------|----------------------------|----------------|-------------------|
| 1    |                 | -0.11166         | -0.08842                   | -0.06429       | -0.08994          |
| 2    | C10             | -0.00161         | -0.08196                   | -0.09311       | -0.09078          |
| 3    | C100            | 0.54328          | -0.12480                   | -0.09976       | -0.11814          |
| 4    | C1000           | 0.63403          | -0.11749                   | -0.05550       | -1.66590          |
| 5    | C2000           | 0.68932          | -0.10778                   | -0.00270       | -5.61643          |
| 6    | C3000           | 0.75908          | -0.09621                   | 0.04892        | -12.01904         |

In SVM regression R2 Value= 0.75908

Hyper Parameter: C3000

Kernel: Linear

## 3.Decision Tree

| S.NO | Criterion      | Splitter | R Value |
|------|----------------|----------|---------|
| 1    | squared error  | best     | 0.69911 |
|      | squared error  | random   | 0.72320 |
| 2    | friedman_mse   | best     | 0.68626 |
|      | friedman_mse   | random   | 0.71263 |
| 3    | absolute_error | best     | 0.67879 |
|      | absolute_error | random   | 0.71472 |
| 4    | poisson        | best     | 0.71379 |
|      | poisson        | random   | 0.68738 |

In Decision Tree Regression R2 Value= 0.72320

#### 4.Random Forest

| S.NO | n_estimators | R Value |
|------|--------------|---------|
| 1    | 10           | 0.83303 |
| 2    | 50           | 0.84983 |
| 3    | 100          | 0.85383 |

In Random Forest Regression R2 Value= 0.85383